

Environmental Product Declaration



in accordance with ISO 14025:2006 and EN 15804:2012+A2:2019/AC:2021 for:

Ductile Iron Pipes of C25 Pressure Class, including Ductile Iron Fittings



from

Electrosteel Castings Limited

Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
Licensee:	EPD India
EPD registration number:	EPD-IES-0020878
Version date:	2025-04-18
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Valid until:	2030-04-17
<i>An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com.</i>	



*EPD of multiple diameter pipe based on the weighted average result of DI pipe including DI fittings



General information

Programme information

Programme:	The International EPD® System	Licensee: EPD India
Address:	EPD International AB Box 210 60, SE-100 31 Stockholm Sweden	EPD India, a licensee of the International EPD® System 422, Midas, Sahar Plaza Mumbai, India-400059
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Accountabilities for PCR, LCA and independent, third-party verification

Product Category Rules (PCR)

CEN standard EN 15804 serves as the Core Product Category Rules (PCR)

Product Category Rules (PCR): *PCR 2019:14 Construction products, version 1.3.4 Published on 2024.04.30. Based on CEN standard EN 15804. CEN standard EN 15804 serve as the core PCR. UN CPC code 41292.*

PCR review was conducted by: The Technical Committee of the International EPD®System. See <https://www.environdec.com/about-us/the-international-epd-system-about-the-system> for a list of members.

Life Cycle Assessment (LCA)

LCA accountability: *Electrosteel Castings Limited.*

LCA and EPD developer: Dr. Rajesh Kumar Singh, Thinkstep Sustainability Solutions Pvt. Ltd., a Sphera Company

Third-party verification

Independent third-party verification of the declaration and data, according to ISO 14025:2006, via:

☒ EPD verification by individual verifier

Third party verifier: Dr. Hüdayi Kara,
Metsims Sustainability Consulting,
4 Clear Water Place, Oxford OX2 7NL, UK
Email: hudai.kara@metsims.com

Approved by: The International EPD® System

Procedure for follow-up of data during EPD validity involves third party verifier:

☐ Yes ☒ No

[Procedure for follow-up the validity of the EPD is at minimum required once a year with the aim of confirming whether the information in the EPD remains valid or if the EPD needs to be updated during its validity period. The follow-up can be organized entirely by the EPD owner or together with the original verifier via an agreement between the two parties. In both approaches, the EPD owner is responsible for the procedure being carried out. If a change that requires an update is identified, the EPD shall be re-verified by a verifier]

The EPD owner has the sole ownership, liability, and responsibility for the EPD.

EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units); have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterization factors); have equivalent content declarations; and be valid at the time of comparison.

For further information about comparability, see EN 15804 and ISO 14025.

Company information

Owner of the EPD: Electrosteel Castings Limited.

Contact: Atindra Narayan Dey

Email: adey@electrosteel.com

Description of the organization:

Electrosteel Castings Limited (ECL) is a well-established company with over six decades of experience in the water infrastructure business. It is one of the largest Ductile Iron (DI) pipe manufacturers in the Indian subcontinent. In 1994, Electrosteel became a pioneer by using state-of-the-art technology to set up the first Ductile Iron Spun Pipe Plant in India. Today, it stands as the world's third-largest producer of spun pipes. The company has a strong brand presence globally, serving a diverse customer base across the Indian subcontinent, Europe, North and South America, Africa, the Middle East, and Southeast Asia. Electrosteel has maintained its technological leadership through continuous product innovation and technical upgrades, focusing on sustainable operation. A widespread marketing network, supported by dedicated professionals, ensures that products are delivered directly to customers' doorsteps.

Product-related or management system-related certifications:

All facilities at Electrosteel Castings Limited (Khardah Works) are ISO 9001, ISO 14001, SA 8000, ISO 45001 & ISO 50001 certified.

Name and location of production site(s):

Electrosteel Castings Limited (Khardah Works)

30, B. T. Road, Khardah, P.O. Sukchar, Dist. 24 Parganas (N), West Bengal, India- 700115



Product information

Product name: Ductile Iron Pipes of C25 Pressure Class, including Ductile Iron Fittings.

Product identification: Electrosteel Castings Limited has state-of-the-art manufacturing facilities for manufacturing Ductile Iron (DI) Pipes & Fittings ranging from DN 80 to DN 1000. The DI Pipes & Fittings are produced in accordance to the standards ISO 2531, EN 545. DI Pipes ranging from DN 700 to DN 1000 are classified under C25 pressure class.

UN CPC code: 41292.

Geographical scope: India.

Product description:

Electrosteel Castings Limited manufactures DI pipes and fittings with various coatings and linings, tailored to the specific conditions of installation and use. The life cycle assessment study encompasses the entire pipe system, including DI pipes (including cement mortar lining), fittings, and accessories like gaskets.



DI Pipe



Example of DI fitting



Standard Gasket

Main characteristics of pipe:

- Used for conveyance of raw water, clear water and wastewater.
- Suitable for fluid temperature between 0° C and 500° C (excluding frost).
- Ductile iron grade- Minimum tensile strength (Rm): 420 MPa / Minimum elongation after fracture=10%.
- Modulus of Elasticity: 1.7×10^{10} kg/m².
- Hardness: Max. 230BHN.
- Density: 7050kg/m³.
- Pipe thickness is designed with a minimum factor of safety of 3.

Internal lining:

- Centrifugally applied cement mortar lining of blast furnace slag cement/Sulphate resistant cement/ High Alumina cement.
- Polyurethane lining.
- Ceramic Epoxy lining.
- Any other lining as agreed with the customers.

External coating:

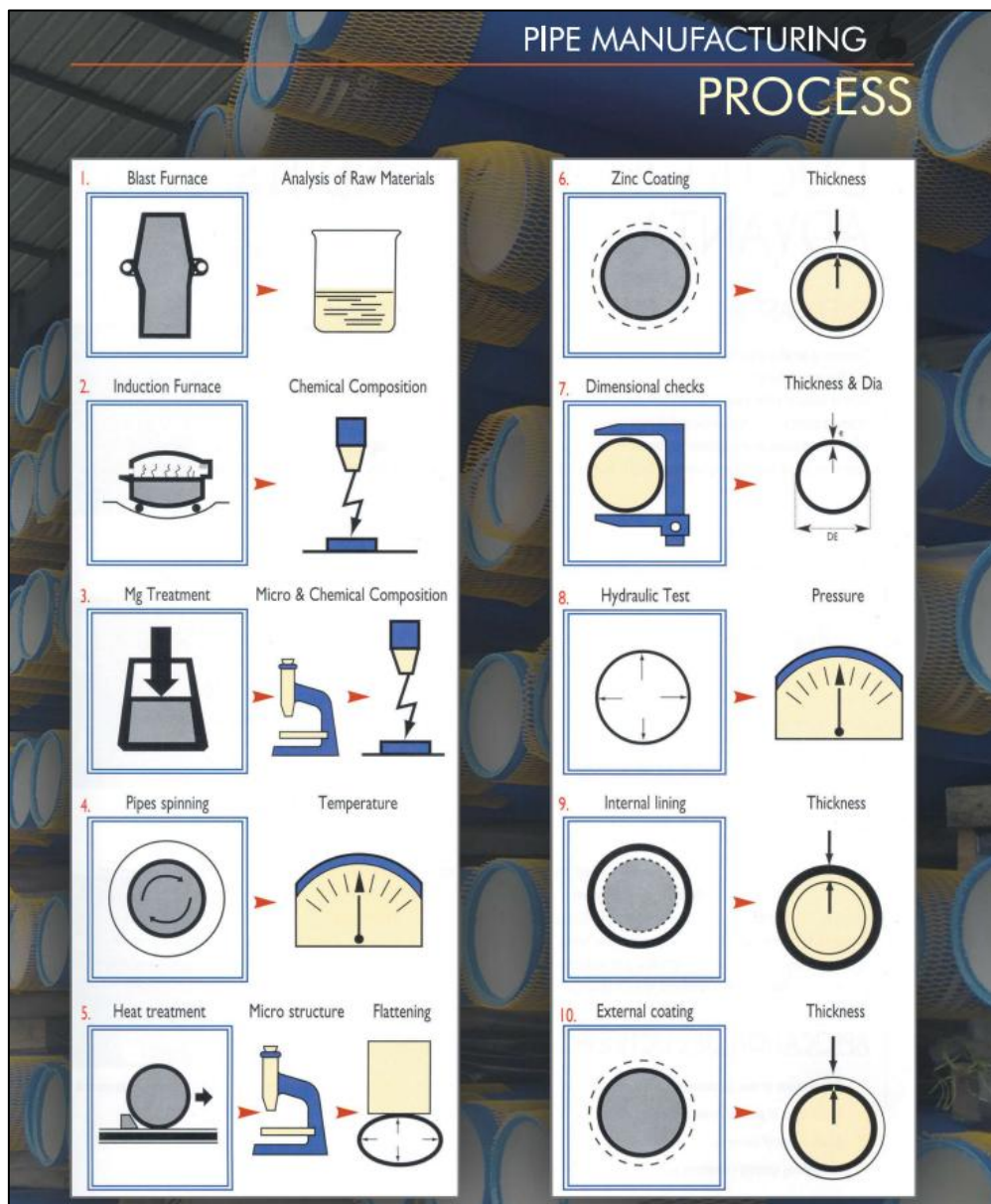
- Metallic Zinc coating having a mass of 130g/m² or 200g/m². The metallic Zinc coating is

covered with a finishing layer of bitumen or Epoxy.

- Zinc Aluminium alloy coating has a mass of 400g/m^2 . The Zinc Aluminium coating is covered with a finishing layer of bitumen or Epoxy.
- Polyurethane coating.
- Any other lining as agreed with the customers.
- Joint: Flexible rubber gasket joint (standard push-on joint, restrained joint and flanged joint).

Manufacturing process:

The manufacturing facility of Electrosteel Castings Limited consists of Sinter plant, Blast Furnace, DRI melting furnace, Induction furnace, DI pipe manufacturing and DI fittings manufacturing units.



Content information

Content declaration of DI Pipes and fittings

Product Components	Weight, kg	Post-consumer recycled material, weight-% of product	Biogenic material, weight-% of product	Biogenic material, kg C/ product or declared unit
Ductile Iron (Pipe)	808.74	0	0	0
Ductile Iron (Fittings)	24.26	0	0	0
Cement mortar lining	155.80	0	0	0
Zinc Coating	3.80	0	0	0
Epoxy / Bitumen Coating	4.20	0	0	0
Gasket	3.20	0	0	0
Total	1000	0	0	0

Products do not contain any substances that can be included in “Candidate List of Substances of Very High Concern for Authorization” and raw materials used are not part of the EU REACH regulation.

LCA information

Functional unit / declared unit: 1 tonne of Electrosteel Castings Limited's Ductile Iron Pipes of C25 Pressure Class, including Ductile Iron Fittings.

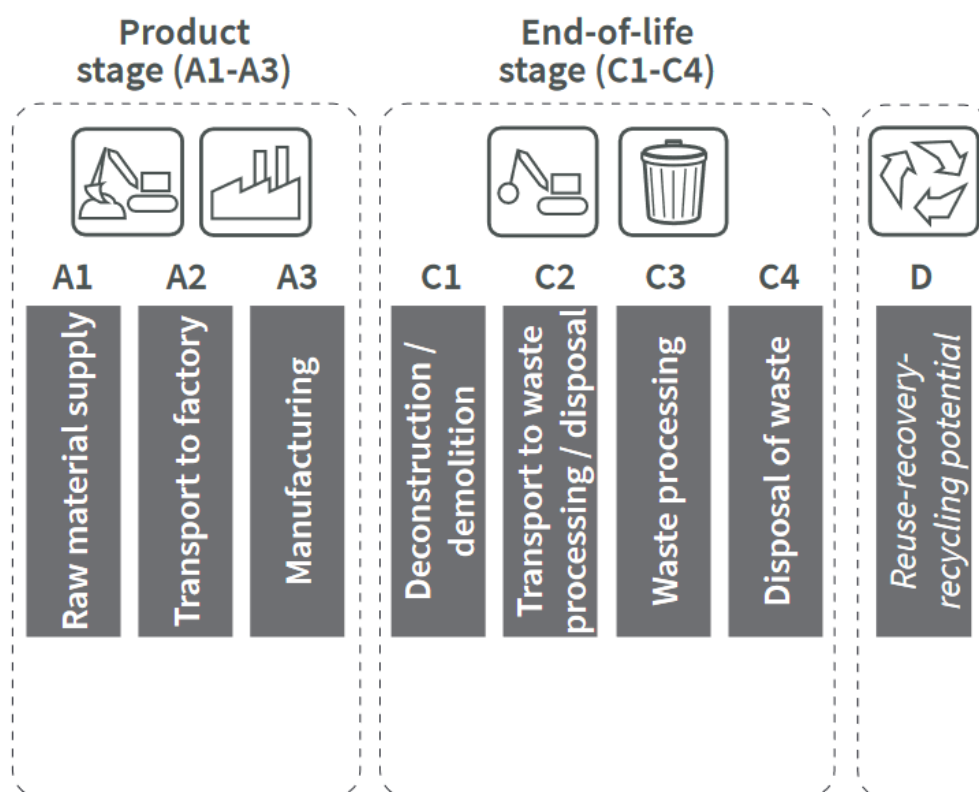
Reference service life: 150 years.

Time representativeness: The collection of foreground data refers to April 2023 to March 2024.

Database and LCA software used: The background data has been taken from the latest available Sphera Managed LCA Content (MLC) 2024.2 and the LCA model was created using Sphera's LCA for Experts, LCA FE (fka GaBi) software, version 10.8.

Description of system boundaries: The system boundary includes cradle to gate stage with additional modules A1-A3, C1-C4, and module D.

Reference package used: EN 15804 reference package based on EF 3.1.



- **Module A1 to A3:**

The product stage includes provision of all materials, products and energy, as well as waste processing up to the end-of waste state or disposal of final residues during the product stage. The electricity in ECL facility has been sourced from a mix of Grid, Captive power plant (CPP), solar photovoltaics and DG sets. The emission factor of the electricity mix for the GWP-GHG indicator is 1.58 kg CO₂eq./kWh. There is no packaging considered in the present assessment.

- **Module C1 to C4:**

Within this EPD, the modules C1-C4 are included. These modules consider the dismantling of the considered product (C1), the transportation of the dismantled components to their End of Life (EoL) destination (C2), the waste processing for recovery or recycling (C3) as well as the disposal (C4), if given. At EoL, the steel (ductile iron) material leaves the product system in C3 for recycling in Module D. The considered EoL scenario for the ductile iron material is 88% recycling and 12% landfill.

Category	Subcategory	Unit	Quantity
Collection process	Collected separately	kg	1000
	Collected with mixed construction waste	kg	0
Recovery	Reuse	kg	0
	Recycling	kg	733
	Landfill	kg	0
	Incineration	kg	0
	Incineration with energy recovery	kg	0
	Energy conversion efficiency rate	kg	0
Disposal	Material for final disposal	kg	267
Transport	Deconstruction site to scrap processing plant	km	0
	Scrap processing plant to site for end of waste	km	50

- Module D:

Module D includes any declared benefits and loads from net flows leaving the product system that have not been allocated as co-products and that have passed the end-of-waste state in the form of reuse, recovery and/or recycling potentials.

Metals are assumed to reach the end of waste status directly at the construction site. The treatment as well as net benefits and loads of reuse or recycling potentials (for the net scrap amount only) are grouped to module D.

In the EoL net scrap approach (Amount of steel recycled at EoL – Scrap input from previous product life cycles) has been used.

Data quality assessment and declaration:

Process	Source type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1- A3
Manufacturing of product	Collected data	EPD owner	2024	Primary data	42.86%
Generation of electricity used in manufacturing of product	Database and collected data	Sphera MLC 2024.2 and EPD owner	2024	Primary data	22.04%
Transport of raw materials to the manufacturing site	Database	Sphera MLC 2024.2	2024	Primary data	4.70%
Total share of primary data, of GWP-GHG results for A1-A3					69.59%

Note: The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that do not capture all relevant aspects of data quality. The indicator is not comparable across product categories.

Cut-off criteria:

The environmental impact of the product studied has been assessed by considering all significant processes, materials, and emissions. Excluded flows are assumed to have a negligible impact, contributing less than 5% to the cumulative impact assessment categories. The production of capital equipment, facilities, and infrastructure required for manufacture has not been considered.

Data quality and sources:

Data quality is compliant with ISO 14025:2006. All primary data were collected for the financial year 2024. All background data come from the Sphera MLC 2024.2.

Allocation:

With any multi-product system, allocation rules are defined to relate the system inputs and outputs to each of the products. Several methods are documented in ISO 14040:2006 and ISO Technical Report 14049. The main co-products for Blast furnace, DI pipe manufacturing are allocated.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results)

	Production			Installation		Use Stage							End-of-Life				Next product system
	Raw material supply	Transport to manufacturer	Manufacturing	Transport to building site	Installation into building	Use / application	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction / demolition	Transport to EoL	Waste processing for reuse, recovery, recycle	Disposal	Reuse, recovery, or recycling
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Modules declared	X	X	X	NR	NR	NR	NR	NR	NR	NR	NR	NR	X	X	X	X	X
Geography	India												GLO	GLO	GLO	GLO	GLO
Specific data used	69.59%			-	-	-	-	-	-	-	-	-	-	-	-	-	-
Variation - products	<10%																
Variation – Sites	0%																

X- Module declared. NR- Not reported.

Results of the environmental performance indicators

The environmental performance for the functional unit of 1 tonne of C25 pressure class DI pipe, including DI fittings of Electrosteel Castings Limited are reported below using the parameters and units as specified in PCR 2019:14 v1.3.4.

The estimated impact results are only relative statements, which do not indicate the endpoints of the impact categories, exceeding threshold values, safety margins and/or risks.

Mandatory impact category indicators according to EN 15804+A2:2019

Results per 1 tonne DI pipe of C25 pressure class, including DI fittings							
Impact indicators	Unit	A1-A3	C1	C2	C3	C4	D
GWP-total	kg CO ₂ eq.	2.53E+03	0.00E+00	4.59E+00	0.00E+00	4.12E+00	-1.27E+03
GWP-fossil	kg CO ₂ eq.	2.53E+03	0.00E+00	4.59E+00	0.00E+00	4.11E+00	-1.27E+03
GWP-biogenic	kg CO ₂ eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
GWP-luluc	kg CO ₂ eq.	1.65E+00	0.00E+00	1.03E-04	0.00E+00	1.07E-02	-1.69E-01
ODP	kg CFC -11 eq.	2.25E-09	0.00E+00	8.61E-14	0.00E+00	1.10E-11	-1.71E-09
AP	Mole of H ⁺ eq.	8.92E+00	0.00E+00	5.25E-03	0.00E+00	2.99E-02	-3.12E+00
EP- freshwater	kg P eq.	1.32E-03	0.00E+00	6.93E-07	0.00E+00	1.60E-05	-2.97E-04
EP- marine	kg N eq.	1.76E+00	0.00E+00	1.76E-03	0.00E+00	7.92E-03	-5.00E-01
EP- terrestrial	Mole of N eq.	1.93E+01	0.00E+00	2.00E-02	0.00E+00	8.67E-02	-4.48E+00
POCP	kg NMVOC eq.	5.47E+00	0.00E+00	5.37E-03	0.00E+00	2.62E-02	-2.03E+00
ADPE	kg Sb eq.	5.83E-03	0.00E+00	5.34E-08	0.00E+00	2.04E-07	-7.21E-03
ADPF	MJ	2.59E+04	0.00E+00	5.93E+01	0.00E+00	5.36E+01	-1.27E+04
WDP	m ³ world equiv.	1.65E+02	0.00E+00	5.56E-03	0.00E+00	4.51E-01	-8.59E+01
Acronyms	Caption: GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - luluc = global warming potential (land use only); ODP = ozone depletion; AP = acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestic = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity.						

Additional mandatory and voluntary impact category indicators

Results per 1 tonne DI pipe of C25 pressure class, including DI fittings							
Impact indicators	Unit	A1-A3	C1	C2	C3	C4	D
GWP-GHG*	kg CO ₂ eq.	2.52E+03	0.00E+00	4.56E+00	0.00E+00	9.13E+00	-1.27E+03
Acronyms	GWP-GHG*= global warming potential (greenhouse gases) This indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide emissions and uptake and biogenic carbon stored in the product with characterization factors (CFs) based on IPCC (2013)						

Resource use indicators according to EN 15804+A2:2019

Results per 1 tonne DI pipe of C25 pressure class, including DI fittings							
Impact indicators	Unit	A1-A3	C1	C2	C3	C4	D
PERE	MJ	1.64E+03	0.00E+00	7.06E-02	0.00E+00	8.46E+00	-5.00E+02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	1.64E+03	0.00E+00	7.06E-02	0.00E+00	8.46E+00	-5.00E+02
PENRE	MJ	2.59E+04	0.00E+00	5.93E+01	0.00E+00	5.36E+01	-1.27E+04
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRT	MJ	2.59E+04	0.00E+00	5.93E+01	0.00E+00	5.36E+01	-1.27E+04
SM	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	m ³	5.85E+00	0.00E+00	1.71E-04	0.00E+00	1.31E-02	-1.29E+02
Acronyms	Caption: PERE = Use of renewable primary energy excluding the renewable primary energy resource used as raw materials; PERM = Use of renewable primary energy as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding the non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

Output flows & Waste Indicators according to EN 15804+A2:2019

Results per 1 tonne DI pipe of C25 pressure class, including DI fittings							
Impact indicators	Unit	A1-A3	C1	C2	C3	C4	D
HWD	kg	2.75E-06	0.00E+00	7.54E-10	0.00E+00	1.32E-08	-9.47E-07
NHWD	kg	3.54E+01	0.00E+00	9.09E-04	0.00E+00	2.66E+02	-1.53E+01
RWD	kg	1.24E-01	0.00E+00	7.44E-06	0.00E+00	5.20E-04	-1.39E-03
CRU	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MFR	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
MER	kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Acronyms	Caption: HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; CRU: Components for re-use, MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy						

Additional environmental performance indicators according to EN 15804+A2:2019

Results per 1 tonne DI pipe of C25 pressure class, including DI fittings							
Impact indicators	Unit	A1-A3	C1	C2	C3	C4	D
PM	Disease incidences	1.43E-04	0.00E+00	5.17E-08	0.00E+00	3.68E-07	-2.92E-05
IR	kBq U235 eq.	1.03E+01	0.00E+00	6.72E-04	0.00E+00	5.74E-02	-2.86E+00
ETF	CTU _e	1.16E+04	0.00E+00	2.67E+01	0.00E+00	3.20E+01	-6.65E+02
HTP-c	CTU _h	1.68E-05	0.00E+00	4.23E-10	0.00E+00	6.72E-10	-5.20E-06
HTP-nc	CTU _h	8.04E-06	0.00E+00	8.80E-09	0.00E+00	3.32E-08	-2.48E-06
SQP	Pt	1.47E+03	0.00E+00	7.07E-02	0.00E+00	1.03E+01	-1.64E+02
Acronyms	Caption: PM = Particulate matter emissions; IR = Ionising radiation, human health; ETF= Eco-toxicity (freshwater); HTP-c = Human toxicity, cancer effects; HTP-nc = Human toxicity, non-cancer effects; SQP = Soil quality potential/Land use related impact.						

References

- General Programme Instructions of the International EPD® System. Version 4.0.
- PCR 2019:14. Construction Products, Version 1.3.4
- Sustainability of construction works - Environmental product declarations - Methodology for selection and use of generic data; CEN/TR 15941:2010
- EN 15804: EN 15804:2012+A2:2019: Sustainability of construction works -Environmental Product Declarations - Core rules for the product category of construction products.
- EN ISO 14025: EN ISO 14025:2011-10 Environmental labels and declarations - Type III environmental declarations - Principles and procedures
- EN ISO 14040: EN ISO 14040:2009-11 Environmental management - Life cycle assessment - Principles and framework
- EN ISO 14044: EN ISO 14044:2006-10 Environmental management - Life cycle assessment - Requirements and guidelines.
- LCA FE: LCA FE Software System and Database for Life Cycle Engineering, Sphera Solution GmbH, Leinfelden-Echterdingen, 2024 (<https://sphera.com/solutions/product-stewardship/life-cycle-assessment-software-and-data/managed-lca-content/>)
- WSI and Eurofer's Co-Product Allocation Methodology 2014 – A methodology to determine the LCI of Steel Industry Co-products.
- World Steel Association- CO₂ Data Collection User Guide, Version 9 (May 2019).

Version History

Original Version of the EPD, 2025-04-18

Revision 1, 2026-03-09

- Reference service life updated to “150 years” from 100 years.
- Table of Diameter wise weight of DI pipes (including fittings) was removed.